Summer Mathematics Learning Notes

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These notes are organized around the subjects I am studying this summer. Each section should grow as a self-contained learning file: goals, definitions, examples, checkpoints, exercises, and connections to the other subjects.

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Author's Preface

Xuran Lyu

These notes are a record of learning in motion: definitions made precise, examples worked until they become familiar, and questions kept close enough to turn into structure. I am writing them as a companion for a summer of analysis, topology, measure theory, functional analysis, elliptic PDE, complex analysis, and the philosophy around mathematical practice.

The aim is not only to collect results, but to preserve the path toward them: what each idea is trying to solve, why the formal language is natural, and how different parts of mathematics quietly talk to one another. The pages that follow should therefore be read as both reference and notebook: clean where possible, exploratory where useful, and always open to revision.

Xuran Lyu

Contents

Notation

Takeaway. Add symbols here as soon as they first appear in the notes. A good notation table should be short enough to scan but complete enough that future reference does not require hunting through the text.

Symbol	Meaning
$\mathbb{R}$	real numbers
$\mathbb{C}$	complex numbers
$\mathbb{N}$	natural numbers
$\mathbb{F}$	either $\mathbb{R}$ or $\mathbb{C}$
X^*	continuous dual space of X
$\|x\|$	norm of x
$x_n \rightharpoonup x$	weak convergence of x_n to x

Learning Template

Each lesson should follow this rhythm:

1. State the learning goals.
2. Give the intuitive picture before the formal definition. In each intuition note, include first-principles thinking: start from the primitive objects, name the problem we are trying to solve, and explain why the definition is forced or natural.
3. Write the formal definition or theorem.
4. Work through at least one example.
5. Add a checkpoint question.
6. End with the takeaway.

Question Log Convention

When a question is answered in these notes, put the answer in the most relevant subject section and mark the date on which the question was asked. If more questions are added to the same section on the same day, continue under the existing date instead of repeating the date.

CHAPTER 1
Real Analysis

First Topics

1. Supremum and infimum; completeness of $\mathbb{R}$.
2. Sequences, subsequences, Cauchy sequences, and completeness.
3. Open and closed subsets of $\mathbb{R}^n$.
4. Continuity, uniform continuity, and compactness.
5. Differentiation in one and several variables.
6. Uniform convergence and interchange of limits.

Learning Goals. This section is for the foundations of rigorous analysis on $\mathbb{R}$ and $\mathbb{R}^n$: limits, continuity, differentiation, integration, convergence of functions, compactness, and the estimates that make later analysis possible.

Core Questions

- What does it mean for a sequence, function, or family of functions to converge?
- Which properties survive under limits?
- How do compactness, completeness, and continuity interact?
- How do estimates turn intuition into proof?

CHAPTER 2
Topology

First Topics

1. Topological spaces and bases.
2. Continuous maps and homeomorphisms.
3. Subspace, product, and quotient topologies.
4. Compactness and local compactness.
5. Connectedness and path connectedness.
6. Separation axioms and metric spaces.

Big Idea. Topology is the language of closeness without necessarily using a formula for distance. This section studies open sets, neighborhoods, convergence, continuity, compactness, connectedness, and quotient constructions.

Core Questions

- What structure is needed to talk about continuity?
- How do open sets encode convergence and local behavior?
- Why is compactness a substitute for finiteness?
- How do different topologies on the same set change the analysis?

Definition 2.1.

Let X be a set. A *topology* on X is a collection τ of subsets of X, called open sets, such that $\varnothing, X \in \tau$, arbitrary unions of sets in τ lie in τ, and finite intersections of sets in τ lie in τ.

Checkpoint. If two topologies live on the same set, how can one topology be "stronger" or "weaker" than the other?

2.1 How to Prove Something Is a Topology

Intuition / First Principles. To prove a collection of subsets is a topology, do not try to understand every open set one by one. Instead, check that the collection survives the three operations a topology is required to survive.

First-principles thinking. A topology is meant to encode stable local information. If many regions are allowed, then their union should also be allowed, because a point only needs to belong to one observed region. If finitely many conditions are imposed at once, their intersection should still be allowed, because we can shrink a neighborhood enough to satisfy all finitely many requirements. The empty set and whole space are the two boundary cases: observing nothing and observing everything.

Proposition 2.2: Topology Proof Strategy.

Let X be a set and let $\tau \subseteq \mathcal{P}(X)$ be a collection of subsets of X. To prove that τ is a topology on X, prove:

(i) $\varnothing \in \tau$ and $X \in \tau$;

(ii) if $\{U_\alpha\}_{\alpha\in A} \subseteq \tau$, then $\bigcup_{\alpha\in A} U_\alpha \in \tau$;

(iii) if $U_1, \dots, U_n \in \tau$, then $\bigcap_{j=1}^{n} U_j \in \tau$.

Remark 2.3. The most common mistake is to prove closure only under finite unions. Topology requires arbitrary unions but only finite intersections. The reason is that infinitely many local restrictions may shrink away all room around a point.

Example 2.4. In the usual topology on $\mathbb{R}$, arbitrary unions are easy: if a point lies in a union, it lies in one open set from the family, so it inherits a small interval from that one set. Finite intersections use the minimum radius: if x has radius ε_1 inside U_1 and radius ε_2 inside U_2, then it has radius $\min\{\varepsilon_1, \varepsilon_2\}$ inside $U_1 \cap U_2$.

■

Takeaway. Topology gives the general vocabulary for continuity and convergence.

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2.2 Topological Spaces and Hausdorff Spaces

Learning Goals. Understand what a topological space is, why open sets are the basic data, and why the Hausdorff condition is the standard separation condition that makes limits behave like they do in metric spaces.

Intuition / First Principles. A topology tells us which subsets count as observable open regions. Once we know the open sets, we know how to talk about neighborhoods, continuity, convergence, closure, density, and compactness. A Hausdorff space is a topological space where distinct points can be separated by disjoint open neighborhoods.

First-principles thinking. Start with only a set X. A bare set has points, but it has no built-in meaning of "near," "continuous," or "convergent." To do analysis, we must add exactly enough structure to say which regions are open around a point. That added structure is a topology. If we also want limits to behave like ordinary limits in $\mathbb{R}^n$, we need a way to distinguish two different points by local observations; this is the Hausdorff condition.

Definition 2.5.

A *topological space* is a pair (X, τ), where X is a set and τ is a collection of subsets of X such that

(i) $\varnothing \in \tau$ and $X \in \tau$;

(ii) arbitrary unions of sets in τ are in τ;

(iii) finite intersections of sets in τ are in τ.

The elements of τ are called *open sets*.

Definition 2.6.

Let (X, τ) be a topological space. A set $U \subseteq X$ is a *neighborhood* of $x \in X$ if there exists an open set $O \in \tau$ such that

$$x \in O \subseteq U.$$

The important point is that topology does not begin with distance. It begins with open sets. If a metric d is available, then the open balls

$$B(x, r) = \{y \in X : d(x, y) < r\}$$

generate a topology. But many useful topologies are not introduced by first writing down a familiar distance.

Example 2.7. Every metric space (X, d) is a topological space: call $U \subseteq X$ open if for every $x \in U$ there exists $r > 0$ such that $B(x, r) \subseteq U$.

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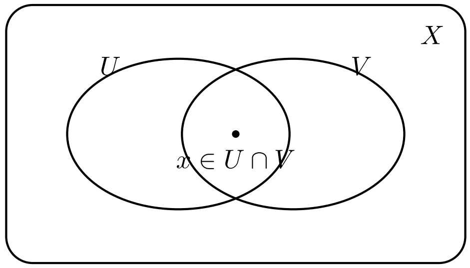

Finite intersections of open sets are open: if $U, V \in \tau$, then $U \cap V \in \tau$.

Figure 1: Open sets are the primitive objects in a topology.

2.2.1 The Usual Topology

Intuition / First Principles. The usual topology is the topology already used in calculus and real analysis, even before being named. On $\mathbb{R}$, it says that open intervals are the basic local regions. On $\mathbb{R}^n$, it says that open balls are the basic local regions.

First-principles thinking. To do analysis on $\mathbb{R}$, we need to say when a point has room around it. The primitive local observation is not the whole line, and not a single point; it is a small interval around the point. A set is open when every point in it has some positive amount of room that still stays inside the set.

Definition 2.8.

The *usual topology* on $\mathbb{R}$ is the topology generated by all open intervals

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Equivalently, $U \subseteq \mathbb{R}$ is open in the usual topology if for every $x \in U$ there exists $\varepsilon > 0$ such that

$$(x - \varepsilon, x + \varepsilon) \subseteq U.$$

Proposition 2.9.

The usual topology on $\mathbb{R}$ is a topology.

Proof. Let

$$\tau_{\text{usual}} = \{U \subseteq \mathbb{R} : \text{for every } x \in U, \text{ there exists } \varepsilon > 0 \text{ with } (x - \varepsilon, x + \varepsilon) \subseteq U\}.$$

We check the three topology axioms.

First, $\varnothing \in \tau_{\text{usual}}$ because there is no $x \in \varnothing$ to check. Also $\mathbb{R} \in \tau_{\text{usual}}$, since for every $x \in \mathbb{R}$ we can take any $\varepsilon > 0$, and then $(x - \varepsilon, x + \varepsilon) \subseteq \mathbb{R}$.

Second, let $\{U_\alpha\}_{\alpha \in A}$ be any family of sets in τ_{usual}. If

$$x \in \bigcup_{\alpha \in A} U_\alpha,$$

then $x \in U_{\alpha_0}$ for some $\alpha_0 \in A$. Since U_{α_0} is open, there exists $\varepsilon > 0$ such that

$$(x - \varepsilon, x + \varepsilon) \subseteq U_{\alpha_0} \subseteq \bigcup_{\alpha \in A} U_\alpha.$$

Thus arbitrary unions are open.

Third, let $U, V \in \tau_{\text{usual}}$. If $x \in U \cap V$, then there exist $\varepsilon_U, \varepsilon_V > 0$ such that

$$(x - \varepsilon_U, x + \varepsilon_U) \subseteq U, \qquad (x - \varepsilon_V, x + \varepsilon_V) \subseteq V.$$

Put $\varepsilon = \min\{\varepsilon_U, \varepsilon_V\}$. Then

$$(x - \varepsilon, x + \varepsilon) \subseteq U \cap V.$$

So $U \cap V$ is open. The same argument, using the minimum of finitely many positive radii, proves that every finite intersection of usual open sets is open. Therefore τ_{usual} is a topology on $\mathbb{R}$. $\square$

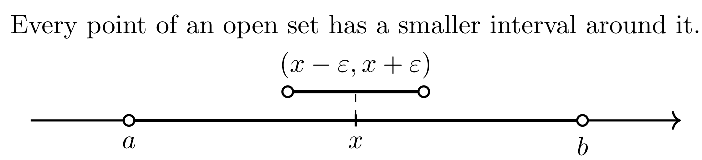

Figure 2: In the usual topology on $\mathbb{R}$, open intervals are the basic neighborhoods.

Proposition 2.10.

A subset $U \subseteq \mathbb{R}$ is open in the usual topology if and only if it is a union of open intervals.

Proof. If U is a union of open intervals, then every point $x \in U$ lies in some interval $(a, b) \subseteq U$. Choose

$$\varepsilon < \min\{x - a, b - x\}.$$

Then $(x - \varepsilon, x + \varepsilon) \subseteq (a, b) \subseteq U$, so U is open.

Conversely, if U is open, then for every $x \in U$ choose $\varepsilon_x > 0$ such that $(x - \varepsilon_x, x + \varepsilon_x) \subseteq U$. Then

$$U = \bigcup_{x \in U} (x - \varepsilon_x, x + \varepsilon_x),$$

so U is a union of open intervals. $\square$

Example 2.11. The sets $(0, 1)$, $(0, 1) \cup (2, 5)$, $\mathbb{R}$, and $\varnothing$ are open in the usual topology on $\mathbb{R}$. ■

Example 2.12. The sets $[0, 1]$, $(0, 1]$, and $\{0\}$ are not open in the usual topology on $\mathbb{R}$. For example, $0 \in [0, 1]$, but every interval $(-\varepsilon, \varepsilon)$ around 0 contains negative numbers, so it is not contained in $[0, 1]$. ■

Definition 2.13.

The *usual topology* on $\mathbb{R}^n$ is the topology generated by Euclidean open balls

$$B(x,r) = \{y \in \mathbb{R}^n : \|x-y\|_2 < r\}, \qquad r > 0.$$

Equivalently, $U \subseteq \mathbb{R}^n$ is open if every $x \in U$ has some Euclidean ball $B(x,r)$ contained in U.

Remark 2.14. In the usual topology, continuity agrees with the ε–δ definition from calculus. A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous exactly when the preimage of every usual open set in $\mathbb{R}^m$ is usual open in $\mathbb{R}^n$.

Common Pitfall. Open does not mean "has no boundary" in an informal visual sense. Open means every point of the set has a full small neighborhood still inside the set. Also, a set can be neither open nor closed, such as $(0,1]$ in the usual topology on $\mathbb{R}$.

Takeaway. The usual topology is the standard topology of real analysis: intervals on $\mathbb{R}$, balls on $\mathbb{R}^n$, and the familiar meaning of limits and continuity.

2.2.2 Common Types of Topologies

Intuition / First Principles. Different topologies on the same set represent different amounts of observable information. A finer topology has more open sets, so it can distinguish points and local behavior more sharply. A coarser topology has fewer open sets, so it sees less.

First-principles thinking. When a topology appears, ask: where did its open sets come from? Sometimes they come from a distance, sometimes from an order, sometimes from a larger ambient space, and sometimes from a requirement that certain maps become continuous. This question usually explains what the topology is designed to measure.

changing topology changes convergence and continuity

coarser $\longrightarrow$ finer

fewer open sets | more open sets

less information | more information

Figure 3: A topology is a choice of observational resolution.

Definition 2.15: Discrete and Indiscrete Topologies.

Let X be a set.

- The *discrete topology* is $\tau = \mathcal{P}(X)$. Every subset of X is open.
- The *indiscrete topology* is $\tau = \{\varnothing, X\}$. Only $\varnothing$ and X are open.

Remark 2.16. The discrete topology is the finest topology on X; it has the most open sets possible. The indiscrete topology is the coarsest topology on X; it has the fewest open sets possible.

Definition 2.17: Metric Topology.

If (X, d) is a metric space, the *metric topology* is the topology whose open sets are those $U \subseteq X$ such that for every $x \in U$ there exists $r > 0$ with

$$B(x, r) = \{y \in X : d(x, y) < r\} \subseteq U.$$

The usual topology on $\mathbb{R}^n$ is the metric topology generated by the Euclidean metric.

Definition 2.18: Cofinite Topology.

Let X be a set. The *cofinite topology* on X is

$$\tau_{\text{cofinite}} = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}.$$

So the nonempty open sets are exactly the sets whose complements are finite.

Example 2.19. On an infinite set X, the cofinite topology is much coarser than the discrete topology. It says that an open set is large if it misses only finitely many points. Any two nonempty cofinite open sets intersect, so an infinite cofinite space is not Hausdorff.

■

Definition 2.20: Subspace Topology.

If (X, τ) is a topological space and $Y \subseteq X$, the *subspace topology* on Y is

$$\tau_Y = \{Y \cap U : U \in \tau\}.$$

Open sets in Y are obtained by cutting open sets of X down to Y.

Example 2.21. The interval $[0, 1]$ inherits a topology as a subspace of $\mathbb{R}$. The set $[0, \frac{1}{2})$ is open in $[0, 1]$ because

$$[0, \tfrac{1}{2}) = [0, 1] \cap (-\tfrac{1}{2}, \tfrac{1}{2}),$$

even though $[0, \frac{1}{2})$ is not open in $\mathbb{R}$ with the usual topology.

■

Definition 2.22: Product Topology.

If X and Y are topological spaces, the *product topology* on $X \times Y$ is generated by products

$$U \times V,$$

where $U \subseteq X$ and $V \subseteq Y$ are open. These rectangles are the basic open sets.

Definition 2.23: Quotient Topology.

Let $q : X \to Y$ be a surjective map from a topological space X onto a set Y. The *quotient topology* on Y is defined by declaring $U \subseteq Y$ open exactly when

$$q^{-1}(U)$$

is open in X.

Remark 2.24. Subspace topology means "inherit open sets from a larger space." Product topology means "make coordinate projections continuous." Quotient topology means "glue points together and keep exactly the open sets whose preimages were open before gluing."

Definition 2.25: Order Topology.

On a linearly ordered set, the *order topology* is generated by open intervals

$$(a, b) = \{x : a < x < b\},$$

together with the appropriate one-sided intervals near endpoints. The usual topology on $\mathbb{R}$ is also its order topology.

Definition 2.26: Weak Topology.

Let X be a set and let $\mathcal{F}$ be a collection of maps $f : X \to Y_f$, where each Y_f is a topological space. The *weak topology generated by* $\mathcal{F}$ is the coarsest topology on X that makes every $f \in \mathcal{F}$ continuous.

Example 2.27. In functional analysis, if X is a normed vector space, the *weak topology* $\sigma(X, X^*)$ is the coarsest topology on X that makes every continuous linear functional $f \in X^*$ continuous. Thus $x_n \to x$ weakly means

$$f(x_n) \to f(x) \qquad \text{for every } f \in X^*.$$

This topology is usually weaker than the norm topology.

■

Definition 2.28: Zariski Topology.

On $\mathbb{R}^n$ or $\mathbb{C}^n$, the *Zariski topology* declares algebraic sets, meaning common zero sets of polynomials, to be closed. This topology is central in algebraic geometry and is much coarser than the usual topology.

Takeaway. The common topologies to recognize first are: discrete, indiscrete, metric/usual, cofinite, subspace, product, quotient, order, weak, and Zariski. The guiding question is always: what information are the open sets designed to preserve?

Definition 2.29.

A topological space X is *Hausdorff* if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \subseteq X$ such that

$$x \in U, \qquad y \in V, \qquad U \cap V = \varnothing.$$

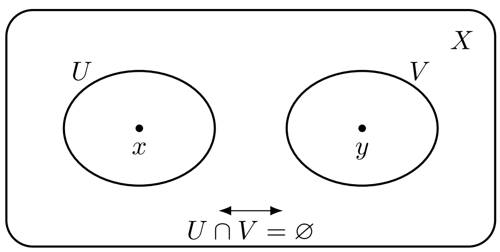

Figure 4: The Hausdorff condition separates distinct points by disjoint open neighborhoods.

Proposition 2.30.

Every metric space is Hausdorff.

Proof. Let (X, d) be a metric space and let $x \neq y$. Then $d(x, y) > 0$. Put

$$r = \frac{d(x, y)}{3}.$$

The balls $B(x, r)$ and $B(y, r)$ are disjoint. Indeed, if some z belonged to both balls, then

$$d(x, y) \leq d(x, z) + d(z, y) < r + r = \frac{2d(x, y)}{3},$$

which is impossible. Thus distinct points can be separated by disjoint open neighborhoods. $\square$

Proposition 2.31.

In a Hausdorff space, a convergent sequence has at most one limit.

Proof. Suppose $x_n \to x$ and $x_n \to y$ with $x \neq y$. Since the space is Hausdorff, choose disjoint open sets U, V with $x \in U$ and $y \in V$. Because $x_n \to x$, eventually $x_n \in U$. Because $x_n \to y$, eventually $x_n \in V$. Therefore, for all sufficiently large n, we would have $x_n \in U \cap V$, contradicting $U \cap V = \varnothing$. $\square$

2.2.3 How to Think About Hausdorff Spaces

Intuition / First Principles. The Hausdorff condition says that points are locally distinguishable. If $x \neq y$, then the topology has enough open sets to build a private open region around x and a private open region around y that do not touch.

First-principles thinking. A topology is a system of tests: open neighborhoods are the local observations we are allowed to make. If two different points can never be separated by disjoint observations, then the topology is too coarse to fully tell them apart. Hausdorffness is the minimum separation condition that restores the analytic behavior we expect from metric spaces: a limit, if it exists, should point to one place rather than two.

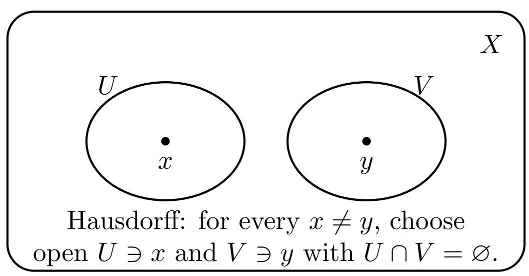

Figure 5: A Hausdorff space can separate any two distinct points.

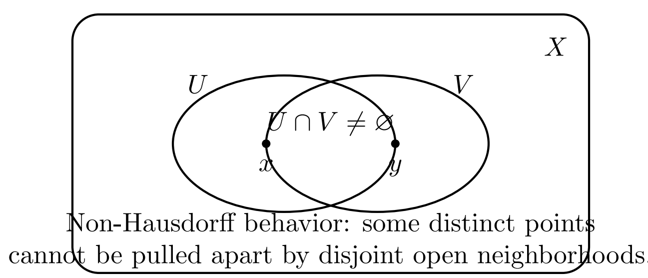

Figure 6: A non-Hausdorff space has points the topology cannot fully distinguish.

Example 2.32. The real line $\mathbb{R}$ with its usual topology is Hausdorff. If $x < y$, set

$$r = \frac{y - x}{3}.$$

Then $(x - r, x + r)$ and $(y - r, y + r)$ are disjoint open neighborhoods of x and y.

■

Example 2.33. Every discrete space is Hausdorff. If $x \neq y$, then $\{x\}$ and $\{y\}$ are disjoint open neighborhoods.

■

Example 2.34. If X has at least two points, the indiscrete topology $\{\varnothing, X\}$ is not Hausdorff. The only open neighborhood of any point is X itself, so two distinct points cannot have disjoint open neighborhoods.

■

Proposition 2.35.

A topological space X is Hausdorff if and only if the diagonal

$$\Delta = \{(x,x) : x \in X\}$$

is closed in the product space $X \times X$.

Proof. Assume X is Hausdorff. If $(x,y) \notin \Delta$, then $x \neq y$. Choose disjoint open sets $U, V \subseteq X$ with $x \in U$ and $y \in V$. Then $U \times V$ is an open neighborhood of (x,y) in $X \times X$, and

$$(U \times V) \cap \Delta = \varnothing.$$

Thus every point outside Δ has an open neighborhood contained in $(X \times X) \setminus \Delta$, so $(X \times X) \setminus \Delta$ is open. Therefore Δ is closed.

Conversely, assume Δ is closed. If $x \neq y$, then $(x,y) \notin \Delta$. Since $(X \times X) \setminus \Delta$ is open, there exist open sets $U, V \subseteq X$ such that

$$(x,y) \in U \times V \subseteq (X \times X) \setminus \Delta.$$

This means $x \in U$, $y \in V$, and $U \cap V = \varnothing$; otherwise a point $z \in U \cap V$ would give $(z,z) \in U \times V \cap \Delta$. Therefore X is Hausdorff. □

Remark 2.36. In very general topology, sequences do not detect all topological behavior. The fully general statement is: in a Hausdorff space, limits of nets are unique. For metric spaces and first-countable spaces, sequences are enough, so the sequence version captures the main analytic intuition.

Remark 2.37. Hausdorffness is one of the reasons topology matters in functional analysis. A topological vector space is usually assumed Hausdorff, so limits of vectors are unique. For a locally convex space, the topology is Hausdorff exactly when continuous linear functionals separate points: for every $x \neq 0$, there exists a continuous linear functional f with $f(x) \neq 0$. In particular, the weak topology $\sigma(X, X^*)$ is Hausdorff precisely when the dual space X^* separates points of X.

2.2.4 Why Hausdorff Spaces Matter

Intuition / First Principles. Hausdorffness matters because analysis depends on being able to identify where a process ends. If a sequence, net, or approximating family can converge to two different points, then the topology is too blurry for many analytic arguments.

First-principles thinking. A limit is supposed to be the final location forced by all sufficiently small observations. If two different points cannot be separated by disjoint neighborhoods, then no local observation can decisively say which point the process is approaching. Hausdorffness says: different points eventually require different local evidence.

1. **It gives uniqueness of limits.** In metric spaces, we take this for granted. In general topology it is a real theorem: Hausdorff separation prevents one convergent process from having two different limits.
2. **It makes compact sets behave like closed bounded sets.** In a Hausdorff space, compact subsets are closed. This is one of the reasons compactness becomes a powerful substitute for finiteness in analysis.
3. **It makes graphs of continuous maps closed.** If Y is Hausdorff and $f : X \to Y$ is continuous, then
$$\operatorname{graph}(f) = \{(x, f(x)) : x \in X\}$$
is closed in $X \times Y$. This matters because closed graphs are a central idea in functional analysis.
4. **It lets functions separate information cleanly.** If X is Hausdorff, points are topologically distinguishable. In functional analysis, this becomes the question: do continuous linear functionals separate vectors?
5. **It rules out pathological identifications.** Non-Hausdorff spaces can intentionally glue points together so tightly that the topology cannot pull them apart. This can be useful in some geometry, but it is usually too weak for ordinary analysis.

Proposition 2.38.

If X is compact and Y is Hausdorff, then every continuous bijection $f : X \to Y$ is a homeomorphism.

Proof. It is enough to prove that f sends closed sets to closed sets. Let $C \subseteq X$ be closed. Since X is compact, C is compact. Since f is continuous, $f(C)$ is compact in Y. Because Y is Hausdorff, compact subsets of Y are closed. Thus $f(C)$ is closed. Therefore f is a closed map, so the inverse map $f^{-1} : Y \to X$ is continuous. □

Takeaway. Hausdorffness is the topological condition that keeps distinct points distinct from the viewpoint of limits. This is why it appears so often in analysis, functional analysis, and PDE: without it, convergence may stop behaving like convergence.

Checkpoint. To test whether a space is Hausdorff, pick two arbitrary distinct points and ask: can the topology produce two disjoint open neighborhoods, one around each point?

Common Pitfall. The Hausdorff condition is not part of the definition of a topological space. Some topological spaces are not Hausdorff. In analysis, however, Hausdorff spaces are often preferred because uniqueness of limits is usually essential.

Takeaway. A topology tells us what open sets are. A Hausdorff topology is one where distinct points can be separated enough that limits are unique.

CHAPTER 3
Measure Theory

First Topics

1. Sigma-algebras and measurable spaces.
2. Measures and outer measures.
3. Lebesgue measure on $\mathbb{R}^n$.
4. Measurable functions and simple functions.
5. Lebesgue integral.
6. Monotone convergence, Fatou's lemma, and dominated convergence.
7. L^p spaces and inequalities.

Learning Goals. This section is for measurable sets, measures, measurable functions, Lebesgue integration, convergence theorems, and L^p spaces. The aim is to build the integration theory used in probability, PDE, and functional analysis.

Core Questions

- Which sets are measurable, and why can not every subset always be measured consistently?
- What does it mean for a function to be measurable?
- When can limits and integrals be exchanged?
- How does measure theory create the spaces $L^p(\Omega)$?

CHAPTER 4
Functional Analysis

First Topics

1. Normed vector spaces and Banach spaces.
2. Hilbert spaces, inner products, and orthogonality.
3. Bounded linear operators.
4. Hahn–Banach theorem.
5. Uniform boundedness, open mapping, and closed graph theorems.
6. Weak and weak-* convergence.
7. Compact operators and spectral theory.

My Story. I am drawn to the philosophy of mathematics, especially mathematical Platonism, and functional analysis feels like the bridge toward that larger picture. *The Road to Reality* by Roger Penrose is one motivation. **Big Idea.** Functional analysis is

linear algebra plus topology plus analysis: the linear algebra of infinite-dimensional vector spaces.

Core Questions

- What does convergence mean in a space of functions?
- Why is completeness important?
- Why are bounded linear maps exactly the continuous linear maps?
- How do weak topologies and dual spaces reveal compactness?

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4.1 How to Read This Chapter

Big Idea. Functional analysis becomes organized once every example is forced to answer the same three questions:

What is the space? What is its topology or norm? What maps act on it?

The chapter should therefore move from spaces, to norms and topology, to operators and functionals.

1. First identify the *objects*: vectors, sequences, functions, or operators.
2. Then choose the *structure*: topology, metric, norm, inner product, or completeness.
3. Then study the *linear maps*: bounded operators, functionals, integral operators, and Green's operators.
4. Finally use special structure, especially Hilbert space geometry, to represent functionals and solve equations.

Takeaway. An example should not appear before the object it is an example of. The running order is: spaces first, norms next, continuity estimates next, then operators, functionals, and Green's formulas.

4.2 Hierarchy of Spaces

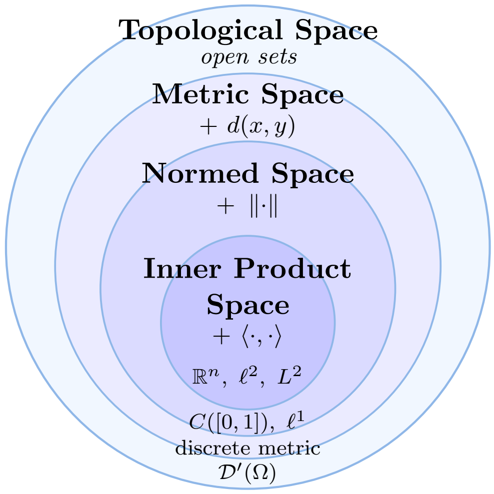
Big Idea. The picture is important because it shows how functional analysis is built by adding structure one layer at a time. Each inner layer carries more information than the outer layer.

Intuition / First Principles. The diagram means "more structure as we move inward." A topological space only knows open sets. A metric space knows distances. A normed vector space knows vector addition, scalar multiplication, and lengths. An inner product space knows angles as well as lengths.

First-principles thinking. Start with the weakest question: what do we need in order to talk about continuity? Open sets are enough, so we get topological spaces. If we also want numerical distances, we add a metric. If we want to add vectors and scale them while measuring their size, we add a vector space structure and a norm. If we want geometry inside the vector space, meaning angles, orthogonality, and projection, we add an inner product. Thus each step answers a stronger question than the previous step.

Proposition 4.1.

Every inner product space is a normed vector space, every normed vector space is a metric space, and every metric space is a topological space.



Containments are strict.

TVS, not metric:
$\mathcal{D}'(\Omega)$ (Schwartz distributions).

Metric, not normed:
discrete metric on $\mathbb{Z}$
(no compatible scalar mult.).

Normed, not inner product:
$C([0,1])$, ℓ^1
(parallelogram law fails).

Figure 7: A hierarchy of spaces. Each ring adds the italicized structure to the one outside it. The label inside each annulus is a canonical space whose richest structure is exactly that layer, so it also witnesses that the containment is strict; the right-hand callout collects the same spaces as named counterexamples.

Proof. If V is an inner product space, define

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

This is a norm, so V becomes a normed vector space. If V is a normed vector space, define

$$d(x, y) = \|x - y\|.$$

This is a metric. If (X, d) is a metric space, define a set $U \subseteq X$ to be open when for every $x \in U$ there exists $r > 0$ such that $B(x, r) \subseteq U$. These open sets form a topology. □

Remark 4.2. The reverse implications are false in general. A topological space need not come from a metric. A metric space need not have any vector space structure. A normed vector space need not come from an inner product. So the diagram should be read as a hierarchy of extra structure, not as a reversible equivalence.

Example 4.3: A Normed Space That Is Not Hilbert. The space $C([0,1])$ with the supremum norm

$$\|f\|_\infty = \sup_{x \in [0,1]} |f(x)|$$

is a normed vector space, hence a metric space and a topological space. But this norm does not come from an inner product. One way to see this is that norms coming from inner products satisfy the parallelogram law,

$$\|f + g\|^2 + \|f - g\|^2 = 2\|f\|^2 + 2\|g\|^2,$$

and the supremum norm does not satisfy this identity in general.

■

Remark 4.4. Function spaces, sequence spaces, and operator spaces are not extra levels between "normed" and "inner product." They are the main kinds of objects that functional analysis studies. Depending on the chosen structure, each one may be a topological vector space, a metric space, a normed space, a Banach space, or sometimes a Hilbert space.

Remark 4.5. For functional analysis, there is another important object nearby: *topological vector spaces*. A topological vector space is not shown as a simple circle inside metric spaces, because many topological vector spaces are not normed and need not be metrizable. The correct relationship is that every normed vector space is a topological vector space, but not every topological vector space comes from a norm.

Takeaway. This hierarchy is one of the main maps for analysis: topology gives continuity, metrics give distance, norms give linear size, and inner products give geometry.

4.3 Function and Sequence Spaces

Big Idea. Before talking about functionals or operators, first identify the space whose points are being studied. In functional analysis, the points are often functions, sequences, or linear maps.

Example 4.6: Function Space. The space $C([0,1])$ is a function space. A point of this space is a continuous function

$$f : [0,1] \to \mathbb{R}.$$

For example, $f(x) = x^2$ is a point of $C([0,1])$.
To talk about distance, we first put structure on this set. Addition and scalar multiplication are defined pointwise:

$$(f+g)(x) = f(x) + g(x), \qquad (af)(x) = af(x).$$

With these operations, $C([0,1])$ is a vector space. Now define the supremum norm by

$$\|f\|_\infty = \sup_{x \in [0,1]} |f(x)|\,.$$

This is a norm: it is nonnegative, it vanishes only when $f = 0$, it is homogeneous, and it satisfies the triangle inequality because

$$|(f+g)(x)| \le |f(x)| + |g(x)|$$

for every $x \in [0,1]$. Therefore $C([0,1])$ is a normed vector space. The norm induces the metric

$$d(f,g) = \|f-g\|_\infty = \sup_{x \in [0,1]} |f(x) - g(x)|\,.$$

So only after choosing this norm does it make sense to say that two functions are close in the uniform distance.

■

Example 4.7: Sequence Spaces. Typical *sequence spaces* include

$$\ell^p = \left\{ x = (x_n)_{n=1}^{\infty} : \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}, \qquad 1 \leq p < \infty,$$

with norm

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p}.$$

These are normed vector spaces, and in fact Banach spaces. The special case ℓ^2 is a Hilbert space because it has the inner product

$$\langle x, y \rangle = \sum_{n=1}^{\infty} x_n \overline{y_n}.$$

■

Example 4.8: More Function Spaces. Typical *function spaces* include $C([0,1])$, $L^p(\Omega)$, and Sobolev spaces $H^k(\Omega)$. These spaces turn functions into points of a vector space. A norm then lets us say when functions are close. For example,

$$\|f\|_\infty = \sup_{x \in [0,1]} |f(x)|$$

on $C([0,1])$ measures uniform closeness of functions.

■

Remark 4.9: Function Space versus Functional. Use *function space* for a space whose points are functions. Use *space of functionals* or *dual space* for a space whose points are scalar-valued maps on another vector space. Functionals come later because they are maps *on* spaces, not the first spaces themselves.

Takeaway. Always locate the point first. In $C([0,1])$ the point is a function; in ℓ^p the point is a sequence; in an operator space the point will be a linear map.

4.4 Norms and Completeness

Intuition / First Principles. A norm is a choice of what "size" should mean. Once a norm is chosen, it automatically gives a distance by

$$d(x, y) = \|x - y\|.$$

Different norms emphasize different features: maximum size, average size, square-integrable energy, derivatives, or operator amplification.

First-principles thinking. In a vector space, subtraction tells us the error $x - y$. A norm is the rule that turns that error vector into a nonnegative number. So choosing a norm is choosing what kind of error matters.

Definition 4.10: Norm.

Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$. A *norm* on V is a map $\|\cdot\| : V \to [0, \infty)$ such that for all $x, y \in V$ and all scalars a,

(i) $\|x\| = 0$ if and only if $x = 0$;

(ii) $\|ax\| = |a| \, \|x\|$;

(iii) $\|x + y\| \leq \|x\| + \|y\|$.

Definition 4.11: Banach Space.

A *Banach space* is a normed vector space that is complete in the metric induced by its norm. Completeness means every Cauchy sequence has a limit inside the same space.

Example 4.12: Euclidean Norm. On $\mathbb{R}^n$, the Euclidean norm is

$$\|x\|_2 = \left(\sum_{j=1}^{n} |x_j|^2 \right)^{1/2}.$$

It comes from the inner product

$$\langle x, y \rangle = \sum_{j=1}^{n} x_j y_j.$$

This norm measures geometric length.

■

Example 4.13: ℓ^p and ℓ^∞ Norms. For a sequence $x = (x_n)$ and $1 \leq p < \infty$, define

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p}.$$

The limiting maximum-size norm is

$$\|x\|_\infty = \sup_{n \geq 1} |x_n| .$$

Thus ℓ^p norms measure summable size, while ℓ^∞ measures uniform boundedness of a sequence.

■

Example 4.14: L^p Norms. For a measurable function f on a measure space Ω and $1 \leq p < \infty$, the L^p norm is

$$\|f\|_{L^p} = \left(\int_\Omega |f(x)|^p \, dx \right)^{1/p}.$$

The case $p = 2$ is especially important because it comes from the inner product

$$\langle f, g \rangle = \int_\Omega f(x)\overline{g(x)}\, dx.$$

■

Remark 4.15: Sup Norm and L^∞ Norm. On $C([0,1])$, the supremum norm is often written like an L^∞ norm:

$$\|f\|_\infty = \sup_{x\in[0,1]} |f(x)|.$$

The reason is that p-norms measure the size of a function by

$$\|f\|_p = \left(\int_0^1 |f(x)|^p\, dx\right)^{1/p},$$

and as $p \to \infty$, this measurement becomes controlled by the largest value of $|f|$. So L^∞ means "the limiting case where only the largest size matters."

Remark 4.16. There is a small but important distinction. For general measurable functions, the L^∞ norm is defined by the *essential supremum*

$$\|f\|_{L^\infty} = \operatorname*{ess\,sup}_{x\in[0,1]} |f(x)|.$$

This ignores what happens on sets of measure zero, because L^p spaces identify functions that agree almost everywhere. For continuous functions on $[0,1]$, the essential supremum agrees with the ordinary supremum, so on $C([0,1])$ we can safely write

$$\|f\|_{L^\infty} = \|f\|_\infty.$$

Example 4.17: Why L^∞ Uses Essential Supremum. Let

$$h(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

As an ordinary bounded function, $\sup_{x\in[0,1]} |h(x)| = 1$. But $h = 0$ almost everywhere, so as an element of $L^\infty([0,1])$,

$$\|h\|_{L^\infty} = 0.$$

This example explains why L^∞ uses essential supremum rather than ordinary supremum. Continuous functions cannot have this kind of single-point spike, which is why the two notions agree on $C([0,1])$.

■

Example 4.18: Sobolev Norm. For PDE, a norm often needs to measure both a function and its derivatives. A typical Sobolev norm is

$$\|u\|_{H^1(\Omega)} = \left(\int_\Omega |u(x)|^2 \, dx + \int_\Omega |\nabla u(x)|^2 \, dx \right)^{1/2}.$$

This norm says that u is small only when both u and its first derivatives are small in L^2.

■

Takeaway. The main norms to recognize are Euclidean norms, ℓ^p norms, L^p norms, supremum norms, L^∞ norms, and Sobolev norms. The operator norm appears after bounded linear operators are introduced, because it measures the size of a map rather than the size of an ordinary vector or function.

4.5 Estimate Toolkit for Continuity

May 26, 2026

Big Idea. Continuity in normed spaces is usually proved by estimates. The same few inequalities explain why norms are continuous, why addition is continuous, why scalar multiplication is continuous, and why bounded linear maps are continuous.

Remark 4.19: Common Triangle Inequality Patterns. The same idea appears in several forms.

$$|a + b| \le |a| + |b| \qquad \text{for real or complex numbers,}$$

$$\|u + v\| \le \|u\| + \|v\| \qquad \text{for vectors in a normed space,}$$

and

$$d(x, z) \le d(x, y) + d(y, z) \qquad \text{for points in a metric space.}$$

The interpretation is always the same: going directly is no longer than going by way of an intermediate object.

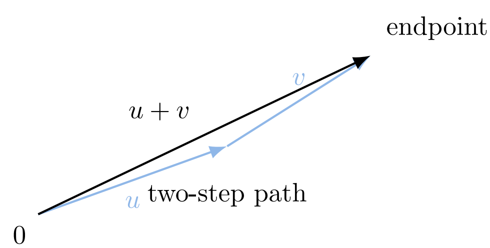


Figure 8: The vector triangle inequality says the direct displacement $u + v$ has length at most the two-step path with lengths $\|u\|$ and $\|v\|$.

Definition 4.20: Lipschitz Map.

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \to Y$ is *Lipschitz* if there exists a constant $L \geq 0$ such that

$$d_Y(f(x), f(y)) \leq L\, d_X(x, y) \qquad \text{for all } x, y \in X.$$

The number L is a Lipschitz constant. Informally, f can stretch distances by at most the fixed factor L.

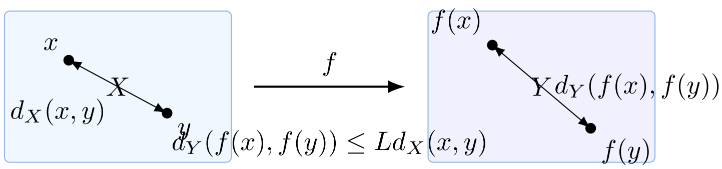


Figure 9: A Lipschitz map has controlled stretching: close inputs remain close outputs, with one fixed constant controlling all pairs.

Proposition 4.21: Lipschitz Implies Continuous.

Every Lipschitz map is continuous.

Proof. Let $f : X \to Y$ be Lipschitz with constant L. If $L = 0$, then f is constant and therefore continuous. If $L > 0$, given $\varepsilon > 0$, choose

$$\delta = \frac{\varepsilon}{L}.$$

Whenever $d_X(x, y) < \delta$,

$$d_Y(f(x), f(y)) \leq L d_X(x, y) < L\delta = \varepsilon.$$

This is exactly the ε–δ definition of continuity. $\square$

Example 4.22: Examples of Lipschitz Maps. The map $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = 3x - 2$$

is Lipschitz with constant $L = 3$, because

$$|f(x) - f(y)| = |3x - 3y| = 3\,|x - y|\,.$$

The sine function is Lipschitz with constant $L = 1$:

$$|\sin x - \sin y| \leq |x - y|\,.$$

This follows from the mean value theorem, since

$$\left|(\sin)'(t)\right| = |\cos t| \leq 1.$$

In any normed space, the norm map

$$x \mapsto \|x\|$$

is Lipschitz with constant $L = 1$, because the reverse triangle inequality gives

$$|\|x\| - \|y\|| \le \|x - y\|.$$

The map $f(x) = x^2$ is not Lipschitz on all of $\mathbb{R}$, because its slope becomes arbitrarily large. But it is Lipschitz on every bounded interval such as $[-R, R]$, since

$$\left|x^2 - y^2\right| = |x - y|\,|x + y| \le 2R\,|x - y|$$

whenever $x, y \in [-R, R]$.

■

Remark 4.23: How to Choose δ in an ε–δ Proof. The goal is always to force

$$d_Y(f(x), f(a)) < \varepsilon$$

from the assumption

$$d_X(x, a) < \delta.$$

The practical method is to estimate the output error in terms of the input error, then choose δ so that the estimate becomes less than ε.
For a Lipschitz map,

$$d_Y(f(x), f(a)) \le L d_X(x, a).$$

Thus it is enough to make

$$L d_X(x, a) < \varepsilon.$$

So choose

$$\delta = \frac{\varepsilon}{L} \qquad (L > 0).$$

This is the cleanest possible case: Lipschitz estimates give δ immediately.

Example 4.24: A Non-Lipschitz Delta Choice: x^2. Prove that $f(x) = x^2$ is continuous at a fixed point $a \in \mathbb{R}$.
Solution. The goal is to show that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|x - a| < \delta \quad \Longrightarrow \quad \left|x^2 - a^2\right| < \varepsilon.$$

Factor the output error:

$$\left|x^2 - a^2\right| = |(x - a)(x + a)| = |x - a|\,|x + a|. \qquad \text{([difference of squares] 1)}$$

The input error $|x - a|$ is visible, but the extra factor $|x + a|$ must be controlled.
Localize x near a by first imposing the preliminary restriction

$$|x - a| < 1. \qquad \text{([local restriction] 2)}$$

Then

$$|x| = |(x-a)+a| \quad \text{([rewrite } x \text{ near } a\text{] 3)}$$
$$\leq |x-a| + |a| \quad \text{([triangle inequality] 4)}$$
$$< 1 + |a|. \quad \text{([use (2)] 5)}$$

Bound the extra factor:

$$|x+a| \leq |x| + |a| \quad \text{([triangle inequality] 6)}$$
$$< 1 + 2|a|. \quad \text{([use (5)] 7)}$$

Return to the output error. Combining (1) and (7),

$$\left|x^2 - a^2\right| = |x-a|\,|x+a| \quad \text{([from (1)] 8)}$$
$$< (1+2|a|)\,|x-a|. \quad \text{([use (7)] 9)}$$

Therefore, to make $\left|x^2 - a^2\right| < \varepsilon$, it is enough to require

$$(1+2|a|)\,|x-a| < \varepsilon. \quad \text{([target inequality] 10)}$$

This is guaranteed by

$$|x-a| < \frac{\varepsilon}{1+2|a|}. \quad \text{([solve for input error] 11)}$$

The two restrictions are (2) and (11), so choose

$$\delta = \min\left\{1, \frac{\varepsilon}{1+2|a|}\right\}. \quad \text{([satisfy both restrictions] 12)}$$

Then $|x-a| < \delta$ implies both $|x-a| < 1$ and $|x-a| < \varepsilon/(1+2|a|)$. Hence (9) gives

$$\left|x^2 - a^2\right| < \varepsilon.$$

Thus x^2 is continuous at a.

■

Remark 4.25: Why the Triangle Inequality Gives Continuity. The triangle inequality does not say "continuity" by itself. It gives an estimate that turns small input errors into a small output error.

For addition, compare two nearby input pairs (x_1, y_1) and (x_2, y_2). The output error is

$$\|(x_1+y_1) - (x_2+y_2)\| = \|(x_1-x_2) + (y_1-y_2)\|.$$

By the triangle inequality,

$$\|(x_1+y_1) - (x_2+y_2)\| \leq \|x_1-x_2\| + \|y_1-y_2\|.$$

This is the same pattern as the familiar real-number inequality

$$|a+b| \leq |a| + |b|.$$

The only change is that the objects being added are vectors. Set

$$u = x_1 - x_2, \qquad v = y_1 - y_2.$$

Then

$$(x_1 + y_1) - (x_2 + y_2) = u + v,$$

so the normed-space triangle inequality gives

$$\|u + v\| \le \|u\| + \|v\|.$$

Thus if x_1 is close to x_2 and y_1 is close to y_2, then the sum $x_1 + y_1$ is close to the sum $x_2 + y_2$. In fact, with the product metric

$$D((x_1, y_1), (x_2, y_2)) = \|x_1 - x_2\| + \|y_1 - y_2\|,$$

addition is 1-Lipschitz:

$$\|(x_1 + y_1) - (x_2 + y_2)\| \le D((x_1, y_1), (x_2, y_2)).$$

By item 4.21, addition is continuous.

Remark 4.26: Scalar Multiplication. Scalar multiplication has one extra issue: both the scalar and the vector can move. The useful decomposition is

$$bx - ay = b(x - y) + (b - a)y.$$

Therefore

$$\|bx - ay\| \le |b| \|x - y\| + |b - a| \|y\|.$$

If (b, x) is close to (a, y), then b is close to a, so b stays bounded, say $|b| \le |a| + 1$ near a. Hence

$$\|bx - ay\| \le (|a| + 1) \|x - y\| + |b - a| \|y\|,$$

and the right-hand side goes to 0 as $b \to a$ and $x \to y$. This is exactly continuity of the map

$$\mathbb{F} \times V \to V, \qquad (a, x) \mapsto ax.$$

Takeaway. Delta choices and continuity proofs usually come from working backward. Estimate the output error, identify what must be made small, and then choose the input restriction that forces it.

4.6 Topological Vector Spaces

Learning Goals. Understand topological vector spaces as vector spaces whose topology is compatible with addition and scalar multiplication. This is the common general setting behind normed spaces, Banach spaces, Hilbert spaces, weak topologies, and many spaces of distributions.

Intuition / First Principles. A topological vector space is a vector space where it makes sense to say that vectors are close, and where the vector space operations respect that notion of closeness. If x_n is close to x and y_n is close to y, then $x_n + y_n$ should be close to $x + y$. If scalars a_n are close to a and vectors x_n are close to x, then $a_n x_n$ should be close to ax.

First-principles thinking. Start with a vector space V. Linear algebra gives two operations: addition and scalar multiplication. But analysis also needs limits, so we add a topology. The topology is useful only if it agrees with the algebra: small changes in inputs should create small changes in sums and scalar multiples. Therefore the definition is forced by requiring the two structure maps

$$(x, y) \mapsto x + y \qquad \text{and} \qquad (a, x) \mapsto ax$$

to be continuous.

Definition 4.27: Topological Vector Space.

Let V be a vector space over $\mathbb{F}$, where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$. A *topological vector space* is a vector space V together with a topology such that the maps

$$V \times V \to V, \qquad (x, y) \mapsto x + y,$$

and

$$\mathbb{F} \times V \to V, \qquad (a, x) \mapsto ax,$$

are continuous.

The first condition says addition is continuous. The second says scalar multiplication is continuous. Together, they mean that the topology and the linear structure are not fighting each other.

$$\begin{array}{ccccccc} V \times V & \xrightarrow{\;+\;} & V & \qquad & \mathbb{F} \times V & \xrightarrow{\text{scalar multiplication}} & V \\ (x, y) & & x + y & & (a, x) & & ax \end{array}$$

Figure 10: A topological vector space requires both structure maps to be continuous.

Example 4.28: Normed Spaces are Topological Vector Spaces. Every normed vector space is a topological vector space. If V is a normed space, the norm defines a metric

$$d(x, y) = \|x - y\|.$$

Addition is continuous because

$$\|(x_1 + y_1) - (x_2 + y_2)\| \le \|x_1 - x_2\| + \|y_1 - y_2\|.$$

Scalar multiplication is continuous because

$$\|a_n x_n - ax\| \le |a_n| \, \|x_n - x\| + |a_n - a| \, \|x\|,$$

so if $a_n \to a$ and $x_n \to x$, then $a_n x_n \to ax$. For the estimate-based explanation of why these inequalities imply continuity, see items 4.25 and 4.26.

■

Example 4.29: Weak Topology. The weak topology on a normed space X is the topology in which $x_n \rightharpoonup x$ means

$$\varphi(x_n) \to \varphi(x) \qquad \text{for every } \varphi \in X^*.$$

This topology is usually weaker than the norm topology, but it is still compatible with vector addition and scalar multiplication. Thus it is a topological vector space topology.

■

Remark 4.30. Many authors include Hausdorffness in the definition of a topological vector space, while others state it separately. In a Hausdorff topological vector space, limits are unique. This is usually the setting wanted in functional analysis.

Big Idea. In a topological vector space, the behavior near every point is controlled by the behavior near 0. This happens because translation $x \mapsto x + a$ is continuous with continuous inverse $x \mapsto x - a$. So neighborhoods of 0 are the basic local objects.

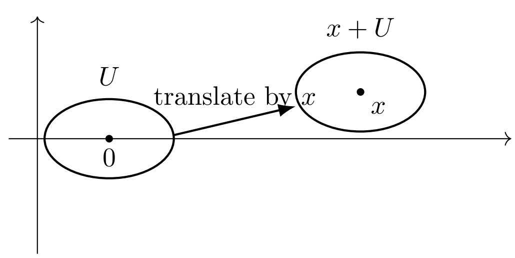


Figure 11: Neighborhoods of any point are translates of neighborhoods of 0.

Checkpoint. Why is a normed vector space automatically a topological vector space?

Solution. The norm gives a metric, hence a topology. The triangle inequality shows that addition is continuous, and the norm estimate for $a_n x_n - ax$ shows scalar multiplication is continuous. □

Takeaway. Topological vector spaces are the natural meeting point of topology and linear algebra. Normed spaces are the most familiar examples, but weak topologies show why functional analysis needs this more general language.

4.7 Operators and Functionals

Big Idea. Once the spaces are clear, the next objects are maps between them. Operators send vectors or functions to other vectors or functions. Functionals are the special case where the output is a scalar.

Definition 4.31: Bounded Linear Operator.

Let X and Y be normed spaces. A linear map $T : X \to Y$ is *bounded* if there exists $C \geq 0$ such that

$$\|Tx\|_Y \leq C\,\|x\|_X \qquad \text{for every } x \in X.$$

The least possible such constant is the *operator norm*

$$\|T\| = \sup_{\|x\|_X \leq 1} \|Tx\|_Y\,.$$

Example 4.32: Operator Norm. If $T : X \to Y$ is a bounded linear operator between normed spaces, its operator norm

$$\|T\| = \sup_{\|x\|_X \leq 1} \|Tx\|_Y$$

measures the largest amount by which T can stretch a vector of size at most 1.

■

Example 4.33: Operator Space. If X and Y are normed spaces, the space

$$\mathcal{L}(X, Y)$$

of bounded linear operators $T : X \to Y$ is an *operator space*. It has the operator norm

$$\|T\| = \sup_{\|x\|_X \leq 1} \|Tx\|_Y\,.$$

The dual space $X^* = \mathcal{L}(X, \mathbb{F})$ is the operator space of continuous linear functionals on X.

■

4.7.1 Functions as Points and Functionals as Points

Intuition / First Principles. The clean distinction is this: in a function space, a point is a function. In a space of functionals, a point is a rule that eats a function or vector and returns a scalar.

First-principles thinking. A mathematical object can play different roles depending on the space we put it in. A function $g \in L^2(\Omega)$ can be a point of the Hilbert space $L^2(\Omega)$. The same g can also generate a functional

$$F_g(f) = \int_\Omega f(x)\overline{g(x)}\,dx.$$

But g and F_g are not literally the same kind of object: g is an input vector, while F_g is a scalar-valued map on vectors.

Example 4.34: Space of Functionals. The dual space $C([0,1])^*$ is a space whose points are continuous linear functionals on $C([0,1])$. For example, for a fixed $a \in [0,1]$, the evaluation map

$$\delta_a : C([0,1]) \to \mathbb{R}, \qquad \delta_a(f) = f(a),$$

is a point of $C([0,1])^*$. It is not a function on $[0,1]$; it is a function whose input is another function.

■

Example 4.35: Integral Functional. If $g \in C([0,1])$, then

$$I_g(f) = \int_0^1 f(x)g(x)\,dx$$

defines a continuous linear functional $I_g \in C([0,1])^*$. Here g is a point of the function space $C([0,1])$, while I_g is a point of the functional space $C([0,1])^*$.

■

Example 4.36: Examples of Integral Functionals. Here are common scalar-valued functionals on a function space:

$$A(f) = \int_0^1 f(t)\,dt \qquad \text{average or total mass,}$$

$$M_1(f) = \int_0^1 tf(t)\,dt \qquad \text{first moment,}$$

and, more generally,

$$I_g(f) = \int_0^1 f(t)g(t)\,dt.$$

In all three examples, the input is a function f, but the output is one number. That is why they are functionals.

■

Remark 4.37: Delta as an Evaluation Functional. The evaluation functional

$$\delta_a(f) = f(a)$$

is not an ordinary integral against a continuous function g. But it can be written formally as

$$\delta_a(f) = \int_0^1 f(t)\,\delta_a(t)\,dt,$$

where δ_a is the Dirac delta distribution concentrated at a. More precisely, it is integration against the Dirac measure:

$$\delta_a(f) = \int_{[0,1]} f(t)\,d\delta_a(t).$$

So the delta symbol is not an ordinary function. It is a way of encoding the functional "evaluate f at a."

Takeaway. The output decides the name. A rule with scalar output is a functional; a rule with vector or function output is an operator.

4.8 Integral Operators and Green's Functions

Big Idea. Integral functionals collapse a function to one number. Integral operators keep a free variable alive, so the output is a whole new function.

Remark 4.38: Fredholm and Volterra Kernels. Fredholm integral equations and Volterra integral equations usually belong first to the world of *integral operators*, not merely integral functionals.
A Fredholm-type kernel $K(x,t)$ defines an operator

$$(Tf)(x) = \int_a^b K(x,t) f(t)\, dt.$$

Here the input is a function f, and the output is again a function of x. Thus

$$T : X \to Y$$

is an operator between function spaces, so T is a point of an operator space such as $\mathcal{L}(X,Y)$ when it is bounded and linear.
A Volterra-type kernel has a variable upper limit:

$$(Vf)(x) = \int_a^x K(x,t) f(t)\, dt.$$

This is still an operator $V : f \mapsto Vf$, because the output depends on x.

Intuition / First Principles. The quick test is: look at the output. If the rule sends a function to a number, it is a functional. If the rule sends a function to another function, it is an operator.

First-principles thinking. A functional collapses all the information in f into one scalar:

$$f \longmapsto \int_0^1 f(t) g(t)\, dt.$$

An integral operator keeps a free variable x alive:

$$f \longmapsto \left(x \mapsto \int_a^b K(x,t) f(t)\, dt \right).$$

So a kernel $K(x,t)$ is like a whole family of functionals, one for each fixed x.

Example 4.39: Fixed x Gives a Functional. For each fixed x, the rule

$$f \longmapsto \int_a^b K(x,t) f(t)\, dt$$

is a scalar-valued functional of f. But collecting all those scalars as x varies gives the function

$$x \longmapsto (Tf)(x).$$

That collection is the integral operator T. This is why Fredholm and Volterra kernels are naturally studied in operator theory, while the single integral $I_g(f)$ above is a functional.

■

Example 4.40: Green's Function for an ODE BVP. Consider the boundary value problem

$$-u''(x) = f(x), \qquad u(0) = u(1) = 0.$$

The Green's function $G(x,t)$ is defined by

$$-\frac{\partial^2}{\partial x^2} G(x,t) = \delta_t(x), \qquad G(0,t) = G(1,t) = 0.$$

For this problem,

$$G(x,t) = \begin{cases} x(1-t), & 0 \le x \le t, \\ t(1-x), & t \le x \le 1. \end{cases}$$

Then the solution is

$$u(x) = \int_0^1 G(x,t) f(t)\, dt.$$

For each fixed x, the rule

$$f \longmapsto u(x) = \int_0^1 G(x,t) f(t)\, dt$$

is a functional of f: it produces one number, the value of the solution at that particular x. The full Green's formula keeps x free:

$$T : f \longmapsto u = Tf, \qquad (Tf)(x) = u(x) = \int_0^1 G(x,t) f(t)\, dt, \qquad 0 \le x \le 1.$$

For the explicit Green's function above, this is

$$u(x) = (1-x) \int_0^x t f(t)\, dt + x \int_x^1 (1-t) f(t)\, dt.$$

This is the integral operator: it sends the input function f to the whole output function u.

■

Intuition / First Principles. Green's functions are the bridge between point sources and operators. The equation

$$L_x G(x,t) = \delta_t(x)$$

says: fix the source point t, hit the system with a unit point source there, and record the response as a function of x.

First-principles thinking. A general forcing $f(t)$ is built by adding many tiny point sources.

The response to a point source at t is $G(x,t)$, so the total response is the superposition

$$u(x) = \int G(x,t)f(t)\,dt.$$

Thus the delta appears in the definition of G, while the final solution formula is an integral operator.

Takeaway. Green's formulas belong in operator theory because the output is the entire solution function u, not merely one scalar value of u.

4.9 Hilbert Spaces and Riesz Representation

Big Idea. Hilbert spaces add inner-product geometry to complete normed spaces. That extra geometry lets every continuous linear functional be represented by inner product with a unique vector.

Theorem 4.41: Riesz Representation for Hilbert Spaces.

Let H be a Hilbert space. For every continuous linear functional $F \in H^*$, there exists a unique vector $g \in H$ such that

$$F(f) = \langle f, g\rangle \qquad \text{for every } f \in H.$$

Remark 4.42. Riesz does not say that every function has a unique functional in every possible function space. It says that in a Hilbert space, every continuous linear functional can be represented uniquely by an inner product with some vector. In $L^2(\Omega)$, this means every $F \in (L^2(\Omega))^*$ has the form

$$F(f) = \int_\Omega f(x)\overline{g(x)}\,dx$$

for a unique $g \in L^2(\Omega)$. So g is a function-as-point, and F is a functional-as-point; Riesz gives a one-to-one correspondence between the two roles.

Remark 4.43: Inner Products and Functionals in Quantum Mechanics. The quantum-mechanical intuition is correct, but the precise statement is this: an inner product is not itself a functional on H, because it takes two inputs,

$$\langle \cdot, \cdot\rangle : H \times H \to \mathbb{C}.$$

It is a scalar-valued form. But if we fix one vector, it becomes a functional of the other vector. In Dirac notation, a ket $|\psi\rangle$ is a vector in the Hilbert space H. The corresponding bra $\langle\psi|$ is the functional

$$\langle\psi| : H \to \mathbb{C}, \qquad \langle\psi|(|\phi\rangle) = \langle\psi, \phi\rangle.$$

So a bra is literally a linear functional on the state space. The number

$$\langle\psi, \phi\rangle$$

is the value of that functional applied to $|\phi\rangle$.

Intuition / First Principles. QM turns the Riesz representation theorem into notation. A state can be viewed as a vector $|\psi\rangle$, but the same state can also be used to test another state by inner product:

$$|\phi\rangle \longmapsto \langle\psi, \phi\rangle.$$

First-principles thinking. A vector contains geometric information. A functional asks a scalar question about an input vector. The inner product is the machine that turns a fixed vector into such a question. In finite dimensions, this is the familiar move from a column vector to a row vector; in Hilbert spaces, Riesz says the same idea still works.

Takeaway. Use *function space* for spaces like $C([0,1])$ and $L^2(\Omega)$, whose points are functions. Use *dual space* or *space of functionals* for spaces like X^*, whose points are scalar-valued maps on X. In Hilbert spaces, Riesz explains why these two worlds can be identified, but they should still be conceptually distinguished.

CHAPTER 5
PDE Analysis: Elliptic PDE

First Topics

1. Laplace and Poisson equations.
2. Weak derivatives and Sobolev spaces.
3. Integration by parts and weak formulations.
4. Lax–Milgram theorem.
5. Dirichlet and Neumann boundary value problems.
6. Maximum principles.
7. Interior and boundary regularity.

Learning Goals. This section is for elliptic partial differential equations, especially weak solutions, Sobolev spaces, boundary value problems, regularity, and variational methods.

Core Questions

- What does it mean to solve a PDE weakly?
- How do boundary conditions enter the function space?
- Why are elliptic equations connected to minimization problems?
- What regularity can be recovered from weak data?

Example 5.1. The Poisson problem

$$-\Delta u = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega$$

is often studied by replacing classical derivatives with weak derivatives and searching for u in a Sobolev space.

■

Checkpoint. Why is the weak formulation of an elliptic PDE often easier to solve than the classical formulation?

Takeaway. Elliptic PDE analysis turns differential equations into problems about function spaces, estimates, and compactness.

CHAPTER 6
Applications

First Topics

1. Heat flow, diffusion, and energy methods.
2. Elasticity and variational principles.
3. Spectral methods and eigenvalue problems.
4. Probability in function spaces.
5. Optimization in Hilbert and Banach spaces.
6. Numerical approximation of PDE.

Learning Goals. This section is for examples where the theory is used: physics, geometry, probability, optimization, numerical methods, data science, and models involving PDE or functional analytic tools.

Core Questions

- Which mathematical structures appear naturally in applications?
- What assumptions make a model analytically tractable?
- How do estimates translate into stability or robustness?
- Which parts of the theory are actually used in computation?

CHAPTER 7
Philosophy of Math

First Topics

1. Platonism, formalism, intuitionism, and structuralism.
2. The role of definitions in creating mathematical worlds.
3. Rigor versus intuition.
4. The meaning of existence theorems.
5. The relationship between computation and proof.
6. The unreasonable effectiveness of mathematics.

Learning Goals. This section is for reflecting on what mathematics is doing: abstraction, rigor, proof, intuition, formalism, mathematical objects, and the relationship between mathematics and the world.

Core Questions

- What is a mathematical object?
- What makes a proof convincing?
- Why does abstraction make mathematics more powerful?
- Is mathematics discovered, invented, or both?
- Why is mathematics so effective in science?

CHAPTER 8
Complex Analysis

First Topics

1. Complex numbers, geometry, and the complex plane.
2. Limits, continuity, and complex differentiability.
3. Cauchy–Riemann equations.
4. Holomorphic functions and analytic functions.
5. Contours, contour integrals, and primitives.
6. Cauchy's theorem and Cauchy's integral formula.
7. Power series, Laurent series, residues, and poles.
8. Conformal maps and harmonic functions.

Big Idea. Complex analysis begins like calculus on $\mathbb{C}$, but complex differentiability is rigid. Once a function is holomorphic, local behavior, power series, contour integrals, and global structure start controlling one another.

Core Questions

- What does it mean for a function $f : \mathbb{C} \to \mathbb{C}$ to be complex differentiable?
- Why is holomorphicity much stronger than real differentiability?
- How does integration around curves reveal information inside a domain?
- Why are singularities classified by local series expansions?
- How do residues compute real integrals and solve applied problems?

Further Reading

Real Analysis

- Walter Rudin, *Principles of Mathematical Analysis.*
- Terence Tao, *Analysis I* and *Analysis II.*

Topology

- James R. Munkres, *Topology.*
- John L. Kelley, *General Topology.*

Measure Theory

- Gerald B. Folland, *Real Analysis.*
- H. L. Royden and P. M. Fitzpatrick, *Real Analysis.*

Complex Analysis

- Lars Ahlfors, *Complex Analysis.*
- John B. Conway, *Functions of One Complex Variable.*
- Elias M. Stein and Rami Shakarchi, *Complex Analysis.*

Functional Analysis

- Erwin Kreyszig, *Introductory Functional Analysis with Applications.*
- John B. Conway, *A Course in Functional Analysis.*
- Walter Rudin, *Functional Analysis.*

Elliptic PDE

- Lawrence C. Evans, *Partial Differential Equations.*
- David Gilbarg and Neil S. Trudinger, *Elliptic Partial Differential Equations of Second Order.*

Applications and Philosophy

- Richard Courant and David Hilbert, *Methods of Mathematical Physics.*
- Eugene Wigner, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences.*
- Stewart Shapiro, *Thinking about Mathematics.*